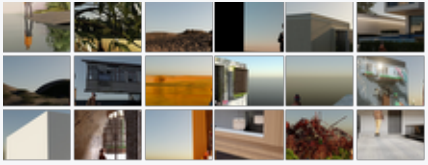
a Assets

BlenderKit

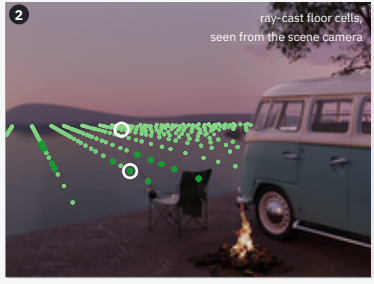


91 indoor & outdoor scenes



102 human subjects · front calibrated on a 12-view turntable

b VLM-guided placement

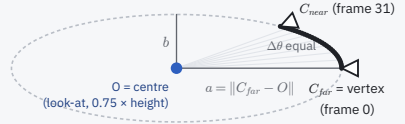


2,054 flat & visible (green, darker = higher score); dropped: 21.6k outside the view, 2,083 hidden, 549 not flat · rings: top-3 kept (re-run without the VLM region)



c Camera trajectories

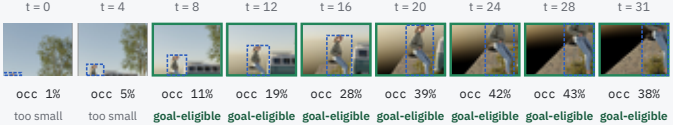
ellipse centred at O with C_far as its vertex; b is solved so it passes through C_near. Endpoints: azimuth ~ U[0°, 360°), distance ~ logU[0.8, 8] m.



rejected (of 393k pairs) **kept 75k · 19 %**

- 152k subject off frame
- 135k camera blocked
- 17k arc midpoint blocked
- 10k endpoints < 15° apart
- 5k degenerate look-at

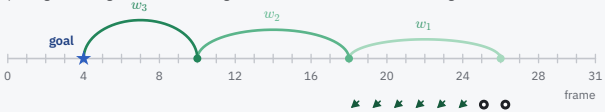
d Render & hindsight label



Each frame is rendered (Cycles, 1024×768) and the subject mesh is projected into it, so its box, bearing, elevation and occupancy are exact — no detector, no annotator. A frame can serve as a goal if occupancy is 10–80 % and ≥ 35 % of the body is visible.

e Windows → action chunks

pick a goal frame g, then starts s with |g − s| ≤ 32; each walks 8 frames toward g



w_3: **9 keyframes (clamped at g)**



a_k: pose change $k \to k{+}1$ in the camera frame, $(\Delta t, R_{6D})$ + shoot, latched once g is reached



8 × 10-D action chunk

≤ 20 windows per trajectory (25 % near goal) · 8 view sectors balanced · 8 val scenes held out

9,889	scene × subject runs	**7,885**	placements (91 scenes)	**393k**	sampled camera pairs	**75k**	trajectories (19 % kept)	**2.4M**	rendered frames	**150k**	training windows